

Problem

Submissions

Leaderboard

Discussions

Given a string S, find the longest border of the string.

A border is a non-empty substring that is both a prefix and a suffix of the string, but it must be shorter than the entire string.

S = "ababa"

Prefixes: "a", "ab", "aba", "abab" Suffixes: "a", "ba", "aba", "baba"

The longest common prefix and suffix is "aba".

Therefore, the answer is:

aba

Input Format

A single line containing the string S.

Constraints

$1 \leq |S| \leq 10^5$ S contains lowercase English letters.

Output Format

Print the longest border of S.

Sample Input 0

ababa

Sample Output 0

aba

Change Theme

Language

Python 3

```
1 # Enter your code here. Read input from STDIN. P
2 s = input().strip()
3 n = len(s)
4 lps = [0] * n
5 length = 0
6 i = 1
7 while i < n:
8     if s[i] == s[length]:
9         length += 1
10        lps[i] = length
11        i += 1
12    else:
13        if length != 0:
14            length = lps[length - 1]
15        else:
16            lps[i] = 0
17        i += 1
18 border_length = lps[-1]
19
20 if border_length > 0:
21     print(s[:border_length])
22 else:
23     print("No border")
24
25
```

Line: 25 Col: 1

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Test case 0

Compiler Message

Test case 1

Success